If we integrate the line element from 0 to $2\pi R$, we get the net magnetic field $\vec{B}$ at point P due to the current-carrying circular loop.

$$\vec{B} = \frac{\mu_0 I}{2} \frac{R^2}{\left(R^2 + z^2\right)^{3/2}} \hat{k}$$

If the circular coil contains N turns, then the magnetic field is

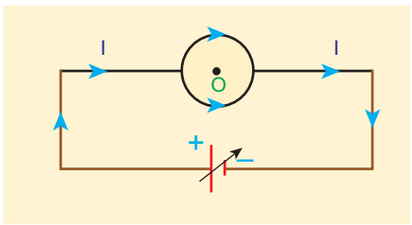
$$\vec{B} = \frac{\mu_0 NI}{2} \frac{R^2}{\left(R^2 + z^2\right)^{3/2}} \hat{k} \qquad (3.37)$$

The magnetic field at the centre of the coil is

$$\vec{B} = \frac{\mu_0 NI}{2R} \hat{k} \quad \text{since } z = 0 \qquad (3.38)$$

EXAMPLE 3.13

What is the magnetic field at the centre of the loop shown in figure?



Solution

The magnetic field due to current in the upper semicircle and lower semicircle of the circular coil are equal in magnitude but opposite in direction. Hence, the net magnetic field at the centre of the loop (at point O) is zero $\vec{B} = \vec{0}$.

3.8.4 Tangent law and Tangent Galvanometer

Tangent galvanometer is a device used to detect very small currents. It is a moving magnet type galvanometer. Its working is based on tangent law (Figure 3.34).



Figure 3.34 Tangent Galvanometer

Tangent law

When a magnetic needle or magnet is freely suspended in two mutually perpendicular uniform magnetic fields, it will come to rest in the direction of the resultant of the two fields.

Let B be the magnetic field produced by passing current through the coil of the tangent galvanometer and B_H be the horizontal component of Earth's magnetic field. Under the action of two magnetic fields, the needle comes to rest making angle θ with B_H, such that

$$B = B_H \tan\theta \qquad (3.39)$$

Construction

Tangent Galvanometer (TG) consists of copper coil of several turns wound on a non-magnetic circular frame. The frame is made up of brass or wood which is mounted vertically on a horizontal base table (turn

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